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ContractNerd: An AI Tool to Find Unenforceable, Ambiguous, and Prejudicial Clauses in Contracts

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Abstract

Contractual agreements often contain clauses that are unfair, creating unjust suffering in one party to the agreement. ContractNerd leverages advanced Large Language Models (LLMs) to analyze contractual agreements and identify issues across four categories: missing clauses, unenforceable clauses, legally sound clauses, and legal but risky clauses. By using a structured methodology that integrates LLM-based clause comparison, enforceability checks against jurisdiction-specific regulations, and assessments of risk-inducing traits, ContractNerd provides a comprehensive analysis of contractual terms. To evaluate the tool's effectiveness, we compare its analyses with those from existing platforms on rental clauses that have led to court litigation. ContractNerd's interface helps users (both drafters and signing parties) to navigate complex contracts, offering actionable insights to flag legal risks and disputes.

Keywords: large language models; generative AI; question answering; contract analysis; rental agreements; employment contracts; linguistics

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1. Introduction

Power imbalances in contractual agreements often result in second parties, notably potential tenants and employees, facing unfair terms. Large corporations, employers, and landlords may use their legal resources to include unreasonable clauses, leaving second parties vulnerable to legal penalties for legitimate actions (e.g., overly restrictive tenant covenants). To address this issue, ContractNerd leverages Large Language Models (LLMs) in the current prototype LLaMA to analyze contractual agreements and alert users to potentially problematic clauses.

The goals are to (i) alert contractual agreement authors to clauses that might be unenforceable or ambiguous and (ii) alert contract recipients to missing clauses or biased clauses. We expect that its use may help avoid unnecessary disputes.

1.1. ContractNerd in Action

ContractNerd can be accessed at https://contractnerd-production.up.railway.app (accessed on 15 October 2025) and ongoing.

The interaction of the reader of a rental agreement proceeds as follows:

1. User starts at the landing page of Figure 1.

2. User uploads a contractual agreement in PDF format on the landing page after selecting a contract type (e.g., rental agreement, employment agreement, etc.) and a jurisdiction (e.g., New York) as in Figure 1.
3. User receives an analysis of the (rental in the provided example) agreement as shown in Figure 2.
   Based on the prompts in Section 4.2, the report flags clauses as missing, unenforceable, legally risky, or legally sound. The goal is to give rental agreement recipients the ability to interpret the results and object to questionable clauses.

Ambiguous Clauses

An example of an ambiguous clause that the system would typically flag can be seen below:

" Tenant must provide written notice of intent to vacate at a reasonable time. "

This is ambiguous because "reasonable" is undefined.

Unenforceable Clauses

Examples of unenforceable clauses would be:

- "Tenant shall not withhold rent for any reason without the landlord's prior written consent."
- "Employee agrees not to work for any business in the United States for two years following termination.": In many jurisdictions—such as California—this is unenforceable under statutes prohibiting broad non-compete agreements. Even in states that allow non-competes, the scope here is excessively broad in geography and duration, making it likely invalid.

Missing Clauses

A missing clause refers to an essential legal or procedural component that is expected in a standard agreement but is absent. For example:

- No language is provided detailing the procedure for returning the tenant's security deposit upon termination of the lease.
- "The agreement contains no procedure for the employee to file grievances or address workplace disputes."

One-sided Clauses

Example of a one-sided clause:

"Tenant shall be responsible for all repairs, regardless of cause."

This shifts unreasonable responsibility onto the tenant, including damage due to structural issues or natural wear and tear, which are typically the landlord's responsibility.

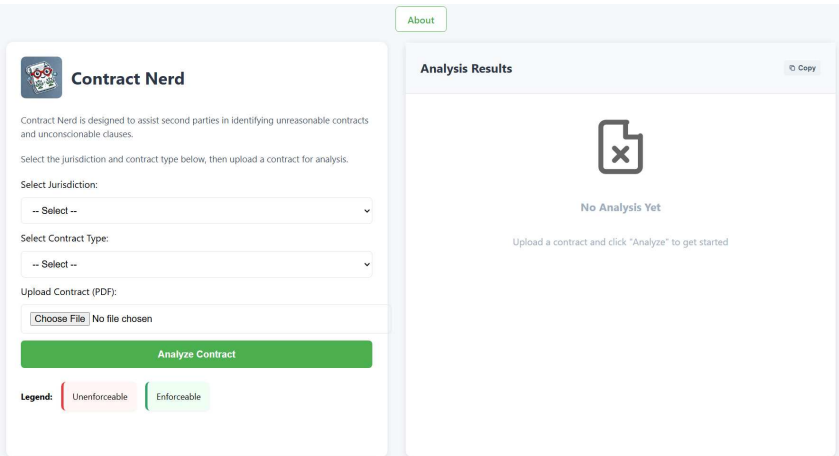

Figure 1. Landing page for ContractNerd user interface.

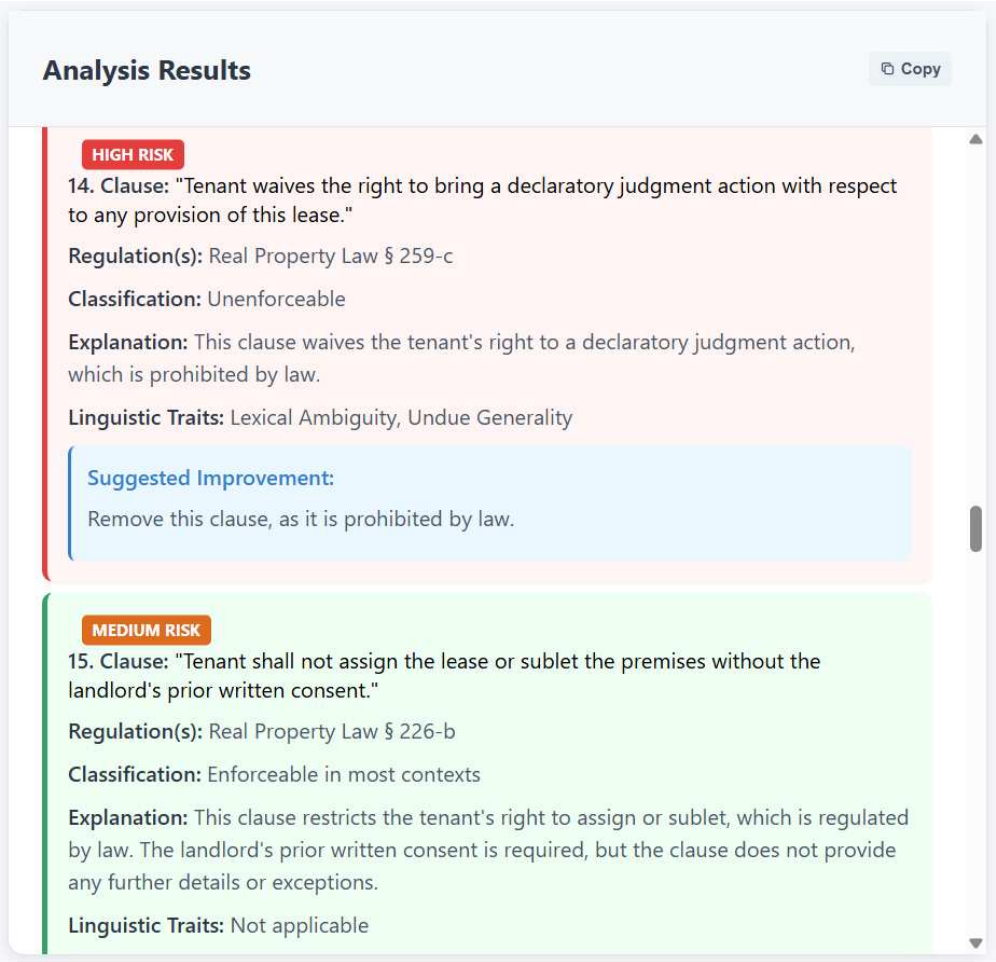

Figure 2. Results displayed after clause analysis.

ContractNerd for rental agreement Writers

Note that the two use cases of ContractNerd are to support both recipients/readers of contract agreements as well as writers of such agreements. Consider again the unenforceable clauses:

"Tenant shall not withhold rent for any reason without the landlord's prior written consent."

To be rendered enforceable, the clause could be rewritten as follows (as ContractNerd suggests):

> "Tenant shall not withhold rent except in cases where the Landlord has breached the warranty of habitability, as provided by law."

1.2. Summary of Contributions

The contributions of this paper and our system ContractNerd are as follows:

1. The presentation of a layman-friendly system to analyze contracts in order to detect biased and/or unenforceable clauses along with explanations.
2. A detailed explanation of the workflow and prompts used to achieve those goals.
3. A set of experiments to evaluate various rental agreement analysis tools, including ContractNerd, based on user feedback and court rulings as detailed by cases highlighted in Thomson Reuters WestLaw.
4. Examples of the application of ContractNerd for employment contracts.
5. A framework for builders of similar document-analysis systems.

2. Related Work

ContractNerd builds on two major research threads: document-level evaluation using Large Language Models (LLMs) and contract-specific legal analysis tools. Our system combines aspects of both while integrating retrieval-augmented generation (RAG) for context-aware clause assessment.

Document-level reasoning using LLMs: Recent work has explored how LLMs can be used to evaluate and reason over long documents. For instance, RAGSys [1] enables real-time self-improvement by storing prior LLM interactions to iteratively enhance retrieval and generation quality. Similarly, DeepRAG [2] separates reasoning and retrieval into distinct phases, supporting multi-hop inference over complex documents.

A wide array of systems now incorporate RAG for specialized applications. QA-RAG [3] targets pharmaceutical compliance, improving accuracy by incorporating structured regulatory content. CaseGPT [4] enhances clinical decision-making by generating medical case reasoning informed by similar past cases. Microsoft's CoRAG [5] takes this further by dynamically reformulating queries in response to intermediate reasoning steps. VideoRAG [6] demonstrates the potential of RAG beyond text by retrieving and integrating multimodal (video) context into LLM outputs.

Contract-Review Systems: There is a growing interest in using LLMs for contract review. Systems like goHeather [7], YesChatContractChecker [8], and Legly [9] automate contract clause identification and enforceability assessment, though often without jurisdiction-aware context. Gao et al. [10] and Rengaraju [11] also analyze rental agreements using LLMs; these works either require model retraining or do not incorporate retrieval-based reasoning.

3. Materials and Methods

3.1. Process Workflow

The processing components, shown in Figure 3, support a three-step interface: ingest, process with prompts, and output.

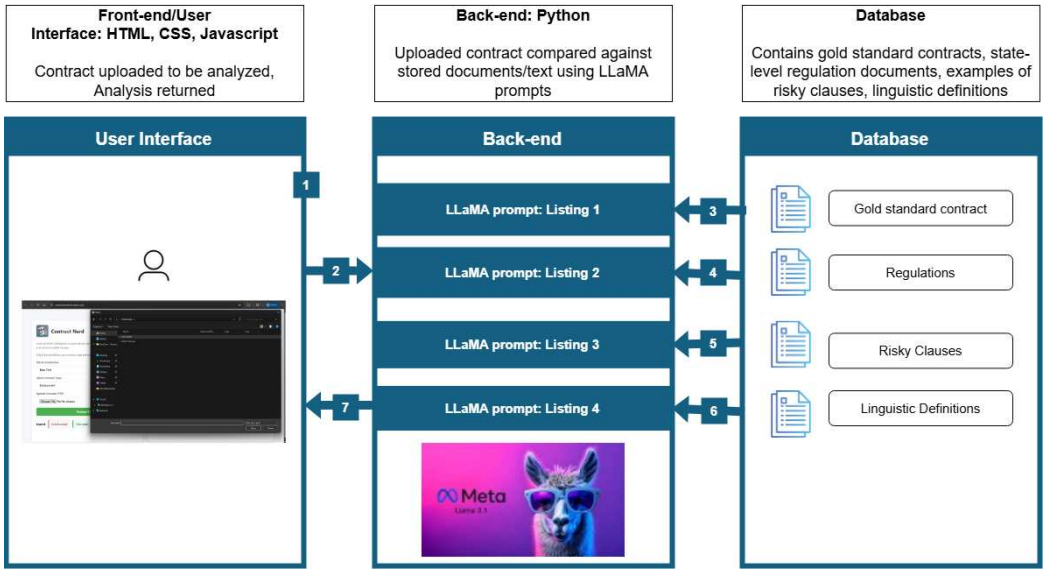

Figure 3. Project workflow consists of three components: (i) ingest document, (ii) apply prompts based on the document and the rules, (iii) deliver output.

3.2. Supporting Data

This project used the following legal document libraries to help in analyzing rental agreements. The following sources provided examples of clauses, regulations, and details of lawsuits:

- Thomson Reuters WestLaw (https://signon.thomsonreuters.com/?productid=CBT&lr=0&culture=en-US&returnto=https%3a%2f%2f1.next.westlaw.com%2fCosi%2fSignOn&tracetoken=0514251349100j4DoQMa9n77sc_TtFHKFtoxdL2lf5mC6WbEDEbRTvUDeMiIF4-T4MoN9gsShPYAfygKn98JnpxOzCEndUOqRDFEM69ODsV7NIaRjeKgogmPjncySCp8PC27VSGx6iTi2aOsdA_OY5HsdxugLvQxGZ52DPBHs6r_SDLiJpUi_MHAyKUOqzn5vs8awNMVVxeD4DFWHDa-RTsBpE_2KPVvr_0FFHTmSapSD37XyNhTNgvK-ULJ7KAI_rCnn30pLg5EB2_yfoxAPKKYaJSNdsLdo28rE04T78jN1nySl6kLlKkD9Sll0JBfKqWtNEoxPZ00zi6oamOWzatGgR7MYg01RXgEKizCQ6dVtfZHz6ni6MV-TeuGU71YYS2S4KHNiVSqk&bhcp=1): A repository offering ongoing access to detailed records of court proceedings, which was consulted during the experimental phase of the study. Last accessed in 14 May 2025.
- Justia (https://contracts.justia.com/contract-clauses/): Serves as a reference for standard rental agreement language, showcasing examples of frequent clauses like "Waiver" etc. Last accessed in 1 November 2024.
- Afterpattern (https://afterpattern.com/clauses): Serves as a reference for standard rental agreement language, showcasing examples of frequent clauses like "Acceleration of Rent" and "Landlord Indemnification.". Last accessed in 1 November 2024.
- Agile Legal (https://www.agilelegal.com/contract-clause-library): Includes a comprehensive library of legal clauses, offering clear explanations of commonly used terms such as "Amendment or Waiver" and "Limitation of Liability." Last accessed in 1 November 2024.
- Sample rental agreements from the official State government websites (https://nystatemls.com/documents/forms/NYStateMLS_Residential_Lease_Agreement.pdf), used to identify key provisions in lease agreements. Last accessed in 1 November 2024.
- PDF documents (https://ag.ny.gov/sites/default/files/tenants_rights.pdf) detailing state-level regulations. Last accessed in 1 November 2024.

4. Implementation

This section starts by discussing the Large Language Model used and its parameter setting. Next, we describe the prompts and their purposes across two main contract types: lease agreements and employment agreements.

4.1. LLM Platform and Hyperparameter Settings

The model's details are specified below:

- **Model**: `LLaMA 3.3 70B`
- **Prompt Strategy**: Zero-shot and few-shot learning
- **Temperature = 0**: Ensures deterministic outputs, vital for structured and accurate clause evaluation.
- **Top_p = 1 and Top_k = 1**: Guarantees the model selects the most probable completion without randomness.
- **Max_tokens = 4096**: Allows for handling large and complex contracts, particularly rental agreements.

4.2. Prompt Strategy

ContractNerd focuses on four principal sources of clause problems:

1. Clauses that are missing but are critical for contract completeness.

2. Clauses that violate local laws that may render a clause unenforceable.
3. Biased clauses that pose risks to the second party and therefore may lead to litigation.
4. Clauses that contain linguistic flaws.

4.3. Missing Clauses

ContractNerd's identification of missing clauses relies on a collection of "complete rental agreements," such as the NY State contract database [12]. Through the prompt in Listing 1 below, the LLaMA model compares the uploaded agreement against this standard contract to flag omissions. Missing clauses are identified using the following prompt:

Listing 1. Prompt used to identify missing clauses in a contract thereby rendering it incomplete.

```
You are a contract compliance analyst comparing a provided contract
against a gold standard contract. Your task is as follows:
1. Identify all clauses in the gold standard contract that are missing
(or materially different) in the provided contract.
2. For each missing/different clause, succinctly explain the legal,
operational, or financial implications of its absence.

Output Format:
Clause: [Exact clause title/text from the gold standard]
Implication if missing: [1--3 bullet points on risks/gaps]

Contract:
_contract.pdf_

Gold Standard Contract:
_gold_standard.pdf_
```

4.4. Enforceability

ContractNerd determines the enforceability of each clause by comparing it against stored regulations.

- The model evaluates each clause's compliance with legal standards and assigns a confidence score (low, medium, high).
- Clauses with high confidence of non-compliance are flagged and reported to the user.
- Clauses with low/medium confidence undergo further scrutiny in subsequent stages.

The LLaMA model prompts use a template to determine compliance with the goal of achieving context-aware and jurisdiction-specific evaluations. The prompt used to compare the agreements against the relevant jurisdiction can be seen below Listing 2:

We designed certain rules in the prompt template to support accurate and legally meaningful assessments:

- **Rule 1: Always include the exact clause text in quotes after "Clause:"**
  This ensures that the model's analysis is anchored to the precise language of the contract, preventing misinterpretation or loss of context that could arise from paraphrasing.
- **Rule 4: Never combine clauses - analyze each separately**
  Separating clauses allows for clause-level risk and enforceability assessment, avoiding false positives or negatives that could result from aggregated or conflated language.
- **Rule 6: Preserve all numbers and names exactly as written**
  Numerical and named details often carry legal significance (e.g., deadlines, amounts,

party identifiers). Preserving them ensures the model evaluates compliance accurately with respect to the intended contract specifications.

The other rules support consistency and formatting, but the above three are particularly important for maintaining the legal integrity and interpretability of the model's outputs.

Listing 2. Prompt to evaluate a contract with respect to regulations to determine compliance.

```
You are a contract language specialist reviewing contract clauses
against regulations.
Output Requirements:
- Analyze each clause individually.
- Use EXACTLY this format for every clause without deviation:

Clause: ``[EXACT FULL CLAUSE TEXT FROM CONTRACT]''
Regulation(s) Implicated: [REGULATION CITATION OR ``None'' IF COMPLIANT]
Reasoning: [CONCISE EXPLANATION OF COMPLIANCE/NON-COMPLIANCE]

Rules:
1. Always include the exact clause text in quotes after ``Clause:''.
2. Never create titles or summarize clauses - use them exactly
as written.
3. If uncertain about compliance, state this in the reasoning.
4. Never combine clauses - analyze each separately.
5. Include all clauses, even compliant ones.
6. Preserve all numbers and names exactly as written.

Contract Clauses:
_contract.pdf_

Regulations:
_laws.pdf_
```

4.5. Riskiness

We define "riskiness" at the clause level using a structured framework grounded in enforceability and legal vulnerability. Each clause is assessed along two dimensions as in Listing 3:

1. **Risk Tier Assignment**:
   - **High Risk**: The clause clearly violates applicable law in commonly occurring scenarios.
   - **Medium Risk**: The clause may raise enforceability concerns depending on the context, jurisdiction, or specific terms of enforcement.
   - **Low Risk**: The clause is generally compliant and enforceable but may still require monitoring due to evolving interpretations or narrow exceptions.
2. **Contextual Enforceability**:
   Each clause is further classified as one of the following:
   - `Enforceable in [specific contexts]`
   - `Unenforceable under [specific conditions]`

This tiered classification aims to provide both a legal and practical assessment of a clause's vulnerability. High-risk clauses tend to be one-sided, overbroad, or contrary to statutory protections; medium-risk clauses are more defensible but may be challenged

under certain factual or procedural circumstances; and low-risk clauses typically align with standard legal practice.

Once the above assessments are completed, ContractNerd conducts a final check to identify clauses that exhibit linguistic flaws that may increase legal uncertainty as shown in Listing 4. The four major sources of such flaws are as follows:

1. **Lexical Ambiguity**: Words allow multiple interpretations without sufficient context (e.g., "reasonable time"), leading to disputes over meaning [13].
2. **Syntactic Ambiguity**: Unclear grammatical structure obscures obligations (e.g., "The contractor will repair the walls with cracks") [14].
3. **Undue Generality**: Provisions are overly broad or vague in scope (e.g., "must meet all necessary standards"), reducing precision and predictability [15].
4. **Redundancy**: Restating obligations unnecessarily, which reduces drafting efficiency and risks internal inconsistency [13].

Addressing these flaws improves enforceability and reduces the likelihood of disputes. In our formulations, we made use of recent research suggesting ways that Large Language Models can assist in detecting such flaws systematically [10].

ContractNerd recommends improvements for medium- and high-risk clauses. These suggestions include alternative phrasing or structural adjustments to improve fairness and enforceability. Low-risk clauses are considered acceptable as written and require no revision.

Listing 3. Prompt to determine regulatory conformance of contract clauses.

```
You are a contract language specialist.
Having compared a contract against a set of regulations, you've
identified the following:
_comparison_

YOUR TASK:
1. Risk Tier Assignment:
 - High Risk: Clearly violates law in common scenarios
 - Medium Risk: Potentially problematic in some contexts
 - Low Risk: Generally compliant but needs monitoring

2. Contextual Classification:
 - For each clause, specify the following:
   * ''Enforceable in [specific contexts]''
   * OR ''Unenforceable under [specific conditions]''
 - Note any exception cases.

3. Improvement Framework:
- For medium/high risk clauses, suggest the following:
  Alternative phrasing options
- For low risk clauses, state the following:
  No changes needed.
```

Listing 3. *Cont.*

```
Output Format:
- Clause: The full text of the clause.
- Regulation(s) Implicated: The exact regulation(s) or criteria
implicated (if unenforceable).
- Classification: Either ‘enforceable‘ or ‘unenforceable‘.
- Risk Tier: [High Risk/Medium Risk/Low Risk]
- Explanation of Classification: A clear explanation of why the
clause was classified as enforceable or unenforceable. If a clause
is one-sided in nature, make mention of this.
- Improvement Guidance: [Actionable steps] You are a contract language
specialist.
```

Listing 4. Analyze contracts from a linguistic point of view.

```
You are provided with a list of contract clauses:
_comparison_
For each clause, the following fields are included:
- Clause:
- Regulation(s) Implicated:
- Classification:
- Risk Tier:
- Explanation of Classification:
- Improvement Guidance:

YOUR TASK:

- For all clauses, reproduce all the provided fields in the output.
- If a clause is classified as unenforceable, analyze it to determine
whether it exhibits any of the following linguistic flaws:
1. Lexical Ambiguity: Words or phrases with multiple interpretations
due to insufficient context (e.g., ’reasonable time’).
2. Syntactic Ambiguity: Grammatical structure is unclear or confusing
(e.g., ’The contractor will repair the walls with cracks’).
3. Undue Generality: The clause is overly broad or vague in scope
(e.g., ’must meet all necessary standards’).
4. Redundancy: The clause repeats the same information unnecessarily.

- Only mention the traits that are actually present in a clause.
If no traits are present, state ’None listed.’
- For enforceable clauses, do not analyze linguistic traits; instead,
simply include all provided fields as-is and state ’Not applicable’
under ’Linguistic Traits Identified.’

The output should specifically be of the following format, with no
exceptions:
- Clause:
- Regulation(s) Implicated:
- Classification:
- Risk Tier:
- Linguistic Traits Identified:
- Explanation of Classification:
- Improvement Guidance:
```

5. Experiments on Rental Contracts

This section details the experiments that were run across two specific types of agreements: rental agreements and employment agreements. The rental agreements Section 5.1 details in-depth analyses to assess the effectiveness of ContractNerd at analyzing lease agreements. The employment agreements Section 6 gives a high-level view of ContractNerd's performance in analyzing employment contracts.

5.1. Rental Agreements

These experiments compare state-of-the-art systems as well as ContractNerd from the points of view of (i) functionality, (ii) performance in flagging unenforceable clauses and assigning risk, (iii) a human evaluation, and (iv) a qualitative expert evaluation. We start with functionality.

5.1.1. Functionality Comparison

To compare state-of-the-art systems, this experiment applied goHeather, Legly, YesChat Contract Checker, and ContractNerd to several rental agreements.

Both ContractNerd and goHeather flagged risky clauses, while YesChat and Legly did not. Sample outputs for an anecdotal test clause are summarized in the prose table below.

As shown in Table 1, we summarize the Clause 1 comparison across the four tools.

Clause 1 (fictitious): "By signing this agreement, the employee agrees to work for the company for the rest of their life, thereby forfeiting their right to resign."

Table 1. How Clause 1 is analyzed by the various tools.

Tool	Tool Analysis Summary
goHeather	Such a clause is likely unenforceable in most jurisdictions, as it violates basic employment rights and principles of personal freedom.
Legly	We found 0 clauses that you need to take a closer look at.
YesChat	This contains several problematic points that could potentially call into question its enforceability: 1. Violation of the Right to Free Choice of Profession 2. Unreasonable Disadvantage 3. Immorality (Section 138 BGB)
ContractNerd	High Risk Regulation(s): Murphy v. American Home Products Corp., 58 N.Y.2d 293 (1983) Classification: Unenforceable Explanation: This case highlights that New York is an "Employment-at-Will" state; this means that an employee may terminate his or her employment at any time. Suggested Improvement: Remove this clause, as it is prohibited by law.

As shown in Table 1, we summarize the Clause 1 comparison across the four tools.

Clause 2 (non-fictitious): "Tenant waives the right to bring a declaratory judgment action with respect to any provision of this lease."

Viewing the results in Table 2, we can see that the different systems support different functionalities. We summarize this in Table 3.

Table 2. Clause 2 analysis comparison by the various tools.

Tool	Tool Analysis Summary
goHeather	Such a clause is likely unenforceable in most jurisdictions, as it violates basic employment rights and principles of personal freedom.
ContractNerd	High Risk Regulation(s): Real Property Law § 259-c Classification: Unenforceable Explanation: This clause waives the tenant's right to a declaratory judgment action, which is prohibited by law. Suggested Improvement: Remove this clause, as it is prohibited by law.

Table 3. System Functionality Comparisons: ContractNerd, goHeather, Legly, and YesChat Contract Checker. A checkmark indicates functionality is provided.

System	Identify Missing Clauses	Give Reason for Unenforceability	Highlight Risky Clauses
GoHeather		✓	✓
Legly			✓
ContractNerd	✓	✓	✓
YesChat		✓	✓

5.1.2. Automated Accuracy Evaluation

This experiment compared these tools using a structured framework comprising the following steps:

1. **Data Used:** Using the keywords in the box below, we extracted 102 clauses from documented rental lawsuit cases in New York State available on Thomson Reuters WestLaw (https://signon.thomsonreuters.com/?productid=CBT&lr=0&culture=en-US&returnto=https%3a%2f%2f1.next.westlaw.com%2fCosi%2fSignOn&tracetoken=0514251349100j4DoQMa9n77sc_TtFHKFtoxdL2lf5mC6WbEDEbRTvUDeMiIF4-T4MoN9gsShPYAfygKn98JnpxOzCEndUOqRDFEM69ODsV7NIaRjeKgogmPjncySCp8PC27VSGx6iTi2aOsdA_OY5HsdxugLvQxGZ52DPBHs6r_SDLiJpUi_MHAyKUOqzn5vs8awNMVVxeD4DFWHDa-RTsBpE_2KPVvr_0FFHTmSapSD37XyNhTNgvK-ULJ7KAI_rCnn30pLg5EB2_yfoxAPKKYaJSNdsLdo28rE04T78jN1nySl6kLlKkD9Sll0JBfKqWtNEoxPZ00zi6oamOWzatGgR7MYg01RXgEKizCQ6dVtfZHz6ni6MV-TeuGU71YYS2S4KHNiVSqk&bhcp=1 (accessed on 1 November 2024)) in which the legal dispute focused on specific contractual terms. We did no further filtering. Each clause was labeled enforceable or unenforceable based on the court's finding of enforceability.
   Here were the keywords used as the basis for the advanced search within WestLaw:

   ```
   landlord, tenant, indemnification clause,
   rental agreement, lease agreement
   ```

2. **Tool Inference:** Each extracted clause was input into the respective tools under evaluation: Zero-Shot ContractNerd, goHeather, Legly, and YesChat.
3. **Performance Measurement:** The outputs of the tools were compared against the ground truth (i.e., the court's ruling) to compute standard classification metrics: accuracy, precision, recall, and F1 score.

As shown in Table 4, ContractNerd achieved the highest overall performance across all metrics compared to the other tools.

Table 4. Tool performance: The positive case is "Unenforceable." Each tool's performance includes 95% confidence intervals and p-values for statistical significance testing against a null classifier (majority class = "Enforceable"). The null model (random classifier) predicts "Unenforceable" with probability $p = \frac{32}{102}$, matching the observed class prevalence. All tools significantly outperform the null ($p < 0.05$), indicated in green. A paired permutation test was conducted on the predictions of ContractNerd against goHeather, revealing a p-value of 0.1796. This suggests that the difference in performance is not large enough to be statistically distinguishable from random variation given the sample size of 102 clauses.

Tool	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
goHeather	74.6	67.2	70.5	68.8
95% CI	(65.7, 82.4)	(41.9, 72.7)	(55.6, 87.1)	(49.3, 75.6)
p-value	<0.05	0.215	0.010	0.108
Legly	61.3	58.2	55.6	56.9
95% CI	(52.9, 70.6)	(40.0, 75.0)	(38.9, 73.5)	(41.3, 70.8)
p-value	<0.05	0.237	0.298	0.298
YesChat	53.9	52.1	49.8	50.9
95% CI	(45.1, 62.7)	(33.3, 69.2)	(32.3, 67.9)	(34.8, 64.7)
p-value	<0.05	0.500	0.570	0.570
ContractNerd	85.3	78.1	75.8	76.9
95% CI	(78.4, 92.2)	(62.5, 91.7)	(60.0, 90.0)	(64.2, 87.3)
p-value	<0.05	<0.05	<0.05	<0.05

5.1.3. Human Feedback Experiment

For a semantically rich comparative evaluation of the best state-of-the-art system goHeather and ContractNerd, we conducted a human evaluation experiment. To set up this experiment, both ContractNerd and goHeather analyzed a set of 26 lease clauses under New York State law. The system outputs were then reviewed side-by-side by human evaluators (both expert and lay) to determine which system provided higher-quality legal analysis.

5.1.4. System Output Collection

ContractNerd and goHeather independently generated structured analyses for each of the 26 clauses. Outputs included clause classification, implicated legal statutes, textual explanation, and any identified linguistic traits.

5.1.5. Evaluation Framework

To assess the quality of each system's output, a comparative evaluation table was developed and provided to six human reviewers (one expert in contracts and five lay users). The lay reviewers comprised a chartered accountant (CIMA), a vice-chair of the institutional review board of a major academic institution, a medical doctor as well as a senior human factors consultant, a lecturer in mechanical engineering, and a college student majoring in finance and economics. Reviewers were instructed to evaluate the two systems' outputs for each clause based on the following three criteria:

1. **Relevance**—How directly the analysis addressed the content and intent of the clause.
2. **Accuracy**—Whether the legal references and interpretations were factually and legally correct.
3. **Completeness**—Whether the analysis covered all significant legal and contextual aspects.

For each clause, reviewers indicated which system produced the better output (or selected "tie" when applicable) and were encouraged to provide optional comments explaining their choice. The experiment named the systems "System A" and "System B" so as

to not bias the reviewers' judgments based on prior knowledge or expectations. The order of presentation (i.e., which system appeared as A or B) was randomized across examples to further reduce potential bias. These precautions were taken to ensure that the evaluation focused solely on the quality of the outputs rather than any preconceived notions about the systems themselves.

Figure 4 shows the results of these tests aggregated to quantify overall comparative performance across the 26 clauses.

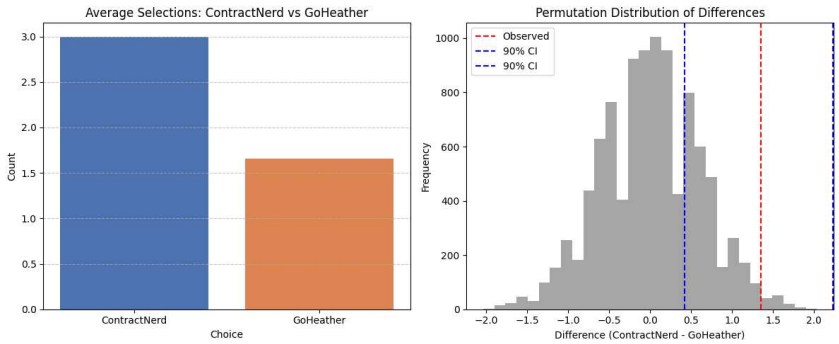

Figure 4. Mean reviewer preferences per question. The paired permutation test (Katari et al., 2021) [16], yielded a *p*-value of 0.009, suggesting the observed preference for ContractNerd is unlikely to arise by chance ($pval < 0.05$). The 90% confidence interval (CI; dashed blue lines) for the mean difference (ContractNerd score − GoHeather score) ranges from 0.42 to 2.23. The histogram shows the permutation distribution (gray) of differences under the null hypothesis, with the observed difference (red dashed line) lying far in its right tail. Results are based on 10,000 resamples of paired responses.

As can be seen from the chart, overall, ContractNerd received more preferential ratings than goHeather.

That said, reviewer commentary not only focused on what ContractNerd does well, but also on potential areas of improvement:

1. "System A seems to give a bit more information but assumes knowledge of legal terms. B explains implications of the various clauses and the output of A could be enhanced by incorporating this language."
2. "System A provides more detail and leaves me feeling more informed (especially under the explanation section). It also mentions the regulation that the clause is under and the classification, providing me with the opportunity to look it up for myself if I want to read about it in more depth. I find this informative."
3. "System B details how this clause may affect the tenant which provides more of a picture of what they would agree to should this clause be part of the tenancy agreement. Though this is not information that causes any legal risk, it is a 'nice-to-have' and is not detailed by System A."
4. "The classification in System A makes the difference - it gives quick judgment without too much reading required. I do however appreciate the additional context given in B."
5. "If the user is a tenant, B offers a better explanation. Perhaps include a toggle to specify landlord/tenant user role"

5.1.6. Expert Qualitative Analysis

As a further form of comparative analysis, Professor Clayton Gillette, a professor of contract law at New York University, conducted a rigorous assessment of both systems' outputs. He evaluated each clause analysis along the three aforementioned dimensions, as we show in Table 5 below and continued in Table A3 in the Appendix B. In general, he found ContractNerd more complete (see, for example, comments regarding points (1) and

(3)) but too, well, nerdy (see comments regarding points (2) and (4)), as shown in Table 5, and continued in Table A3 in the Appendix B.

Table 5. Lease Agreement Clause Analysis by law professor Clayton Gillette.

Clause	System	Analysis	Comment
Tenant shall be liable for early termination fees as specified	ContractNerd (System A)	• Risk Level: Low • Regulation(s): None • Classification: Enforceable • Explanation: This clause makes the tenant liable for early termination fees, which is a standard lease provision. • Suggested Improvement: No changes needed.	Reasonable termination fees may be difficult to know in advance, so requiring that they be detailed in the manner required by B could lead to incomplete specification.
	GoHeather (System B)	• Risk Level: High • Explanation: The clause references early termination fees but does not specify the amount or calculation method. This lack of specificity is risky for the tenant. • Suggested Change: Replace with detailed language.	
Tenant shall not withhold rent for any reason without the landlord's prior written consent.	ContractNerd (System A)	• Risk Level: High • Regulation(s): Real Property Law § 235-b • Classification: Unenforceable under certain conditions • Explanation: This clause restricts the tenant's right to withhold rent, which is regulated by law. The tenant has the right to withhold rent if the landlord breaches the warranty of habitability. • Suggested Improvement: Consider adding language to specify the conditions under which rent withholding is allowed, such as "Tenant shall not withhold rent except in cases where the Landlord has breached the warranty of habitability, as provided by law."	A will be confusing to the average tenant who does not know what the warranty of habitability or "law" provides. The "suggested change" in B does not change the initial language.
	GoHeather (System B)	• Risk Level: High • Explanation: This clause is problematic. Under New York law, tenants have the right to withhold rent if the landlord fails to provide essential services or maintain habitable conditions. This clause attempts to waive that right, which is unenforceable. • Suggested Change: Remove the following: "Tenant shall not withhold rent for any reason without the Landlord's prior written consent."	

Table 5. *Cont.*

Clause	System	Analysis	Comment
No pets allowed	ContractNerd (System A)	• Risk Level: Medium • Regulation(s): NYC Admin. Code § 27-2009.1(b); Westchester County Laws, Chapter 695.11 • Classification: Enforceable in most contexts, Unenforceable under disability laws • Explanation: This clause prohibits pets in the rental unit, which is allowed by law. However, there are exceptions for service animals and emotional support animals, which are not mentioned in this clause. • Suggested Improvement: Consider adding language to specify the exceptions for service animals and emotional support animals, such as "No pets allowed, except for service animals and emotional support animals as required by law."	Specifying the allowance for service animals avoids any issues about the subject.
	GoHeather (System B)	• Risk Level: Low • Explanation: A strict no-pets policy is standard but can be restrictive. In New York City, if a tenant openly keeps a pet for three months and the landlord does not object, the no-pet clause may become unenforceable under the 'three-month rule.'	
Tenant shall be responsible for all attorney fees incurred due to breach of this lease agreement	ContractNerd (System A)	• Risk Level: Medium • Regulation(s): Real Property Law § 234 • Classification: Enforceable in most contexts • Explanation: This clause makes the tenant responsible for attorney fees, which is allowed by law. However, the law also provides that the tenant has a reciprocal right to recover attorney fees if the landlord breaches the lease. • Suggested Improvement: Consider adding language to specify the reciprocal right to recover attorney fees, such as "Tenant shall be responsible for all attorney fees incurred due to breach of this lease agreement, and Landlord shall be responsible for all attorney fees incurred due to breach of this lease agreement by Landlord."	B is shorter and easier to comprehend.

Table 5. *Cont.*

Clause	System	Analysis	Comment
Tenant shall be responsible for all attorney fees incurred due to breach of this lease agreement	GoHeather (System B)	• Risk Level: High • Explanation: This is a high-risk clause. It makes the tenant liable for all attorney fees if they breach the lease, but does not provide for reciprocal rights if the landlord breaches. In New York, such clauses must be reciprocal to be enforceable. • Suggested Change: Replace the following: "Tenant shall be responsible for all attorney fees incurred due to breach of this lease agreement." with "In the event of any legal action arising from this lease, the prevailing party shall be entitled to reasonable attorney fees."	

5.1.7. Other Jurisdictions

The examples detailed above focus on ContractNerd's analysis of clauses that can be found in rental agreements with specific reference to New York state rental laws. However, the approach can be generalized across various jurisdictions, which we show in detail in Appendix A where we apply the same methodology to New Jersey and California. The main takeaways are as follows:

- Table A2, Appendix A shows the different ways in which New Jersey and California balance landlord rights to recover costs and tenant protections under lease cancellation scenarios. In both cases, ContractNerd classifies clauses governing the application of re-renting income toward landlord expenses and tenant debts as medium risk and enforceable, provided that expenses are reasonable and well-documented (see Table A2, Appendix A). However, California law explicitly requires landlords to mitigate damages by making reasonable efforts to re-rent and imposes stricter demands for documentation and expense limits, highlighting a stronger tenant protection framework compared with New Jersey's approach. This suggests that, while both states protect landlord interests, California limits recoverable costs more rigorously.
- Another notable contrast concerns the warranty of habitability. In the case of New Jersey, a high risk classification was assigned to clauses that ambiguously limit the landlord's obligation to provide essential services, deeming such provisions unenforceable if basic habitability standards are not met (Table A2, Appendix A). In contrast, California's approach views similar clauses as low risk, recognizing that the warranty covers only fundamental residential fitness without guaranteeing additional amenities. This difference highlights California's more permissive stance toward landlords' limitations on service levels, whereas New Jersey seemingly prioritizes safeguarding tenant access to essential services under its Residential Landlord-Tenant Act.

These examples show that ContractNerd performs appropriately different analyses of the same contract clauses in different jurisdictions.

6. Employment Agreements

Table 6 below highlights ContractNerd's ability to flag specific regulations in a jurisdiction-specific way in employment law. As we will see, different jurisdictions embody divergent regulatory philosophies regarding employee protection, contractual enforceability, and the permissible scope of employer control.

Table 6. Analysis of Employment Clauses by ContractNerd.

Clause	New Jersey	California
"In the event you terminate your employment with the Company by your resignation for any reason within eighteen (18) months of receiving a Company-paid benefit such as training, tuition reimbursement, seminars, compensation/performance based bonuses, relocation bonus/expenses, un-recovered advances against future salary, initial airfare, etc., the Company reserves the right to recover from you the full cost of such Company-paid benefit and to set off the cost of all or any portion of such Company-paid benefit against any amounts owed by the Company to you upon such termination of employment."	**Risk Tier:** High Risk **Regulation(s):** New Jersey Wage and Hour Law (NJWHL) **Classification:** Enforceable **Linguistic Traits:** Not applicable **Explanation:** This clause describes the repayment terms for company-paid benefits, which are subject to the NJWHL. The clause is one-sided in nature, as it only benefits the employer, inflicting significant financial obligations on the employee with no reciprocal requirement on the employer. **Improvement:** Limit covered benefits to a clearly defined list or "including but not limited to" with a defined scope; add pro-rata repayment; require itemized statement of costs; include "to the extent permitted by law" for any set-off; carve out repayment where resignation is due to company breach, constructive dismissal, or legal prohibition.	**Risk Tier:** High Risk **Regulation(s):** Cal Labor Code §221, Cal. Lab. Code § 2802 **Classification:** Enforceable **Linguistic Traits:** Undue Generality **Explanation:** Under California law, wage deductions are strictly regulated. Section 221 prohibits employers from recouping wages unless explicitly permitted by statute or a valid agreement. While recovery of certain expenses (e.g., training costs) may be enforceable if reasonable and tied to actual expenditures, the clause's broad scope risks violating Section 221. **Improvement:** Specify exact repayment terms (e.g., prorated based on time remaining). Exclude wages from set-off and limit recovery to direct, documented costs.
"Employee agrees during the term of Employment not to engage in, work for, assist or be associated with in any way, or have any interest in, any business which competes with Itlize, or any business which solicits customers and markets similar products in those areas in which Itlize currently does business."	**Risk Tier:** High Risk **Regulation(s):** New Jersey Trade Secrets Act (NJTSA) **Classification:** Enforceable **Linguistic Traits:** Undue Generality **Explanation:** The clause restricts the employee from being connected to any competing business in any form during employment. While in-term non-competes are generally enforceable in NJ, the scope here is overbroad (could include passive stock ownership) and vague in geographic reach. "Any interest in" could unintentionally capture harmless investments. **Improvement:** Narrow to competitive activities materially related to the employee's role; exclude passive investments under a certain ownership percentage; define "competes" and "areas" with specificity; link restriction to protection of confidential information or legitimate business interests.	**Risk Tier:** High Risk **Regulation(s):** Cal. Bus. & Prof. Code § 16600 **Classification:** Unenforceable **Linguistic Traits:** Undue Generality **Explanation:** California's Section 16600 voids non-compete agreements unless they fall under statutory exceptions (e.g., sale of a business). This clause's blanket restriction on working for competitors or holding any interest in a competing business during employment is likely unenforceable, as it far exceeds the narrow bounds of permissible restraints. Courts routinely strike down such broad language as an unlawful restriction on lawful profession, trade, or business. **Improvement:** The clause should be narrowed to protect only legitimate business interests, such as trade secrets or confidential information. Remove phrases like "any interest in" and limit restrictions to direct competition or solicitation of specific clients.
"Employee hereby agrees that his/her employment with the Company may be terminated at any time and for any reason, or for no reason."	**Risk Tier:** High Risk **Regulation(s):** New Jersey Law Against Discrimination (NJLAD) **Classification:** Enforceable **Linguistic Traits:** Not applicable **Explanation:** The clause accurately states NJ's at-will employment doctrine but omits recognition of legal limits—employers cannot terminate for discriminatory, retaliatory, or otherwise unlawful reasons. The lack of such carve-outs can create risk of perceived overreach or employee. **Improvement:** Add explicit language that termination rights are "subject to applicable law" and cannot be for discriminatory, retaliatory, or otherwise unlawful reasons; clarify that at-will status can only be modified in writing by authorized representatives.	**Risk Tier:** Medium Risk **Regulation(s):** Cal. Lab. Code § 2922 **Classification:** Enforceable **Linguistic Traits:** Not applicable **Explanation:** California law presumes at-will employment under Section 2922, allowing termination without cause. However, exceptions exist (e.g., violations of public policy, implied contracts, or anti-discrimination laws). **Improvement:** Ensure no contradictory language exists elsewhere in the contract. Explicitly state that the at-will provision cannot be modified except in writing signed by an authorized Company representative.

Table 6. *Cont.*

Clause	New Jersey	California
"The Employee agrees that if the Employee terminates this Agreement prior to the Termination Date or Term Date, the Employee will pay Itlize a sum of One Thousand Dollars ($1000.00) for every month remaining on the Term Date or Termination Date, as a liquidated damages sum for the breach of this provision only, which is in addition to any other damages Itlize may seek for a breach of any other provision in this Agreement including but not limited to, the non-solicitation and non-competition provisions of this Agreement."	**Risk Tier:** High Risk **Regulation(s):** New Jersey Law Against Discrimination (NJLAD) **Classification:** Enforceable **Linguistic Traits:** Not applicable **Explanation:** NJ courts will enforce liquidated damages only if they are a reasonable estimate of probable loss at the time of contracting—not a penalty. Here, $1000/month may be arbitrary and punitive, especially when combined with "in addition to any other damages." **Improvement:** Tie damages to demonstrable costs incurred by the employer; remove "in addition to any other damages" or clarify that total recovery will not exceed actual losses; add carve-outs for termination due to employer breach or mutual agreement; ensure repayment mechanism complies with wage deduction laws.	**Risk Tier:** High Risk **Regulation(s):** Cal. Bus. & Prof. Code § 16600 **Classification:** Unenforceable **Linguistic Traits:** Undue Generality **Explanation:** Liquidated damages must reasonably estimate actual harm, not function as a penalty. Courts may view the monthly fee as punitive, especially if unrelated to demonstrable losses. Additionally, tying this provision to unenforceable non-compete terms (under Section 16600) further undermines its validity. **Improvement:** Remove references to non-compete clauses and recast liquidated damages as reimbursement for actual costs. Use a sliding scale (e.g., prorated repayment based on time worked) and provide a rationale for the amount

Various themes emerge as particularly noteworthy:

- **Divergence in Recovery of Employer Costs**: The first clause—authorizing the employer to recover the cost of company-paid benefits upon employee resignation—illustrates a sharp jurisdictional contrast. In New Jersey, while the clause is broadly enforceable under the New Jersey Wage and Hour Law, its breadth creates a high risk of overreach, particularly when applied without proportionality or itemization. In California, the same clause risks violating strict statutory prohibitions on wage recoupment under Labor Code §221. This demonstrates how California adopts a more employee-centric approach, prioritizing wage integrity over contractual flexibility, whereas New Jersey permits broader enforcement provided statutory procedures are followed.
- **Non-Compete and Restraint-of-Trade Clauses.** The second clause, which functions as an in-term non-compete, can be enforceable in New Jersey under the New Jersey Trade Secrets Act if narrowly drafted. However, the language here is excessively broad—extending even to passive stock ownership—and is therefore vulnerable to legal challenge. In contrast, California law, through Business and Professions Code §16600, makes such restrictions almost entirely unenforceable unless a narrow statutory exception applies. This difference illustrates California's near-total rejection of restraints on employee freedom, compared with New Jersey's conditional acceptance based on reasonableness and legitimate business interests.

7. Limitations

Scalability:

A potential limitation of the current prototype is efficiency when processing very large contractual documents. As contracts increase in length and clause count, a single forward pass through the language model may become impractical due to context window constraints and latency.

To address this, the system can be extended with a *partition-and-union* approach. The algorithm proceeds as follows:

1. **Partitioning:** The document is segmented into smaller, overlapping chunks (e.g., 3–5 pages, or 10–15 clauses per chunk) to ensure that clause boundaries are preserved.
2. **Clause Extraction:** Each chunk is independently analyzed by the language model to identify and classify clauses.

3. **Union of Results:** The results from all chunks are combined into a unified set of clauses. Duplicate or overlapping clauses are merged.
4. **Completeness Tracking:** Metadata (clause IDs, page/line numbers) is maintained during chunking so that all original clauses are accounted for in the final union.
5. **Post-Processing:** The unified clause set is reconciled against the original document to ensure coverage, flagging any missing or unprocessed clauses for re-analysis if necessary.

This strategy would allow the system to scale to longer documents while maintaining clause completeness and minimizing the risk of omission. Such an approach has been successfully applied in related applications (e.g., large-text summarization), and represents a natural extension of the current work.

Considerations of Bias:

As the review paper [17] discusses, machine learning systems are based on data and are therefore vulnerable to the bias inherent in that data. That is a possible consideration for this work for the following reasons:

- Employment contracts, especially as enforceable documents, are strongly anchored in the formal economy. Underground economy work is often carried out without any documentation and certainly no notion of legal enforcement. Therefore, the bias here is that this work is relevant only to those fortunate enough to be in the formal economy.
- Another source of bias is that our work is specific to the United States with an Anglo-Saxon style of laws based on precedents. Because we infer possible questionable clauses based on the analysis of laws, we believe, but haven't proved, that similar methods could apply to countries which follow a rule-based legal system.

Considerations of Accuracy:

While ContractNerd is designed to be advisory rather than authoritative, the high quality of its output might convince users that it is in fact authoritative. That is not so. Partly because of the inexactitude of AI technology and partly because of the irrationality of human actors in the judicial system, we can imagine a scenario in which ContractNerd states that a clause should be permissible, but that a contract drafter loses a lawsuit based on that clause. We urge contract drafters to use ContractNerd as one of many tools that they might use to improve the quality and fairness of contracts.

8. Conclusions

ContractNerd integrates jurisdiction-aware legal retrieval with `LLaMA 3.3`'s reasoning capabilities to analyze different styles of contracts (here, rental and employment) using the same prompt. ContractNerd compares favorably with state-of-the-art tools (goHeather, Legly, and YesChat) in citing enforceable regulations (85.3% accuracy) and flagging risky terms (76.9% F1-score). Three key innovations drive this effectiveness:

1. **Context-Aware Evaluation**: ContractNerd grounds its analysis in specific regulations (e.g., NYS rental laws), reducing false positives in enforceability judgments.
2. **Structured Risk Framework**: The tiered risk classification (High/Medium/Low) and contextual enforceability labels provide both an evaluation and suggestions for rewording, while linguistic trait analysis (lexical ambiguity, undue generality) further helps contract writers refine clause wording.
3. **User-Centric Design**: Human evaluations resulted in a quite dramatic and statistically significance preference for ContractNerd's outputs compared with the state of the art. This held particularly for ContractNerd's regulatory citations and classification clarity.

To see the details of the grading and suggestions that ContractNerd performed across other jurisdictions as well as details of the expert qualitative evaluation, please see the appendices.

Future work centers around overcoming the scalability limitations and expanding the scope of our analysis to different kinds of contracts along with extensive user testing.

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Appendix A. Applying ContractNerd to Rental Agreement Other Jurisdictions

While the examples given above focus on New York, ContractNerd expands across multiple jurisdictions in the USA. Below are examples of clause analyses for leases based in other states in the USA.

Table A1. Lease clause analysis summary.

Clause	New Jersey	California
"Pursuant to this provision, the tenant is to indemnify the landlord 'for any damage flowing from such use'."	**Risk Tier:** Medium Risk **Regulation(s):** N.J.S.A. 2A:18-61.1 **Classification:** Enforceable in specific contexts (e.g., tenant's use directly causes damage) **Linguistic Traits:** Not applicable **Explanation:** May relate to landlord's right to recover damages under Anti-Eviction Act, but language requires clarification. **Improvement:** Rephrase to clarify scope: "The tenant shall indemnify the landlord for damages caused by the tenant's negligence or willful misconduct."	**Risk Tier:** Medium Risk **Regulation(s):** California Civil Code §§ 2782, 2782.5 **Classification:** Enforceable in specific contexts **Linguistic Traits:** Undue Generality **Explanation:** California law limits indemnity clauses in leases, especially if they require the tenant to indemnify the landlord for landlord's own negligence. The phrase "any damage flowing from such use" is broad and ambiguous, potentially exposing the tenant to excessive liability. **Improvement:** Clarify and limit indemnity obligations to damages caused by tenant's negligence or misconduct. Include clear carve-outs for landlord's own negligence or willful misconduct. Use precise language defining "damage" and scope of indemnity.

Table A1. *Cont.*

Clause	New Jersey	California
"If Landlord breaches said agreement the parties hereby agree to use the American Arbitration Association to resolve their disputes and the prevailing party will be allowed to recover costs, expenses, and damages of said arbitration including but not limited to attorneys fees."	**Risk Tier:** Medium Risk **Regulation(s):** N.J.S.A. 2A:18-61.1, N.J.S.A. 46:8-48 **Classification:** Enforceable where arbitration agreement is fair/reasonable **Linguistic Traits:** Not applicable **Explanation:** Subject to Anti-Eviction Act and Truth in Renting Act; fee recovery provisions may be regulated. **Improvement:** Clarify fairness: "Parties agree to AAA arbitration, each bearing own costs unless ruled otherwise."	**Risk Tier:** Low Risk **Regulation(s):** California Arbitration Act (Cal. Code Civ. Proc. §§ 1280–1294.2) **Classification:** Enforceable **Linguistic Traits:** Not applicable **Explanation:** California generally enforces arbitration clauses and mutual attorney's fees provisions unless unconscionable or violating public policy. Specifying the AAA and fee-shifting to the prevailing party aligns with California law and promotes efficient dispute resolution. **Improvement:** Specify procedures for initiation of arbitration, timelines, and confidentiality terms for completeness. Consider defining "prevailing party" to avoid disputes over fee recovery.

Table A2. Lease clause analysis summary continued.

Clause	New Jersey	California
"If this Lease is canceled, or Landlord takes back the Apartment, the following takes place: … (3) Any rent received by Landlord for the re-renting shall be used first to pay Landlord's expenses and second to pay any amounts Tenant owes under this Lease. Landlord's expenses include the costs of getting possession and re-renting the Apartment, including, but not only reasonable legal fees, brokers fees, cleaning and repairing costs, decorating costs and advertising costs."	**Risk Tier:** Medium Risk **Regulation(s):** N.J.S.A. 2A:18-61.1, N.J.S.A. 46:8-48 **Classification:** Enforceable if expenses are reasonable/documentedif expenses are reasonable/documented **Linguistic Traits:** Not applicable **Explanation:** Procedures must comply with Anti-Eviction Act and Truth in Renting Act. **Improvement:** Specify, "Landlord shall apply re-renting income first to reasonable documented expenses, then tenant debtsProcedures must comply with Anti-Eviction Act and Truth in Renting Act. **Improvement:** Specify, "Landlord shall apply re-renting income first to reasonable documented expenses, then tenant debts."	**Risk Tier:** Medium Risk **Regulation(s):** California Civil Code §§ 1951.2, 1951.4 **Classification:** Enforceable under certain conditions **Linguistic Traits:** Not applicable **Explanation:** California requires landlords to mitigate damages by making reasonable efforts to re-rent. Prioritizing expenses before tenant obligations is common but must be reasonable and supported by documentation. Overly broad expense categories may invite disputes over what costs are recoverable. **Improvement:** Define "reasonable" more precisely or provide caps where possible. Require itemization of expenses to be charged against rent proceeds.
"The prong of the warranty of habitability implied in every residential lease by which the landlord covenants that the premises are fit for the uses reasonably intended by the parties does not encompass the level of services and amenities that tenants reasonably expect to be provided under the financial and other terms of their individual leases."	**Risk Tier:** High Risk **Regulation(s):** N.J.S.A. 2A:42-84 **Classification:** Unenforceable if essential services are withheld **Linguistic Traits:** Undue Generality **Explanation:** Warranty of habitability under NJ Residential Landlord-Tenant Act requires essential services. **Improvement:** State explicitly, "Landlord shall provide [list essentials] per warranty of habitability requirements."	**Risk Tier:** Low Risk **Regulation(s):** California Civil Code §§ 1941, 1941.1 **Classification:** Enforceable **Linguistic Traits:** Undue Generality **Explanation:** The warranty of habitability requires landlords to provide premises fit for basic residential use but does not guarantee amenities or service levels beyond those implied by law or explicitly stated in lease agreements. This clause correctly limits landlord obligations consistent with California law. **Improvement:** Consider specifying which services or amenities are excluded or included to avoid ambiguity.

Appendix B. Expert Qualitative Analysis Continued

Table A3. Lease Agreement Clause Analysis by law professor Clayton Gillette continued.

Clause	System	Analysis	Comment
Late fee: Fixed Amount. $50 for each occurrence	ContractNerd (System A)	• Risk Level: Medium • Regulation(s): Real Property Law § 238-a • Classification: Enforceable in most contexts • Explanation: This clause charges a late fee of $50, which is allowed by law. However, the law also requires that the late fee not exceed 5% of the monthly rent or $50, whichever is less. • Suggested Improvement: Consider adding language to specify the conditions under which the late fee is charged, such as "A late fee of $50 or 5% of the monthly rent, whichever is less, shall be charged for each occurrence."	ContractNerd is easier to understand. The reference to New York law raises more issues than it resolves for the average reader. Both clauses are relevant and accurate.
	GoHeather (System B)	• Risk Level: High • Explanation: A fixed late fee is standard, but in New York, late fees are capped at $50 or 5% of the monthly rent, whichever is less. If the rent is less than $1000, this clause may violate New York law. • Suggested Change: Replace "Late fee: Fixed Amount. $50 for each occurrence" with "Late fee: $50 or 5% of the monthly rent, whichever is less, for each occurrence, in accordance with New York law."	
Tenant waives the right to bring a declaratory judgment action with respect to any provision of this lease.	ContractNerd (System A)	• Risk Level: High • Regulation(s): Real Property Law § 259-c • Classification: Unenforceable • Explanation: This clause waives the tenant's right to a declaratory judgment action, which is prohibited by law. • Suggested Improvement: Remove this clause, as it is prohibited by law.	ContractNerd is better, though I am not sure it is accurate to say that it is prohibited. Tenants cannot be forced to waive, but if they agree, the clause may be enforceable.
	GoHeather (System B)	• Risk Level: High • Explanation: This is a high-risk clause. Waiving the right to seek a declaratory judgment limits the tenant's ability to challenge the legality or enforceability of lease provisions in court. Such waivers are generally disfavored and may be unenforceable under New York law, as tenants cannot be forced to waive statutory rights. • Suggested Change: Remove "Tenant waives the right to bring a declaratory judgment action with respect to any provision of this lease."	

Table A3. *Cont.*

Clause	System	Analysis	Comment
Tenant shall not assign the lease or sublet the premises without the landlord's prior written consent	ContractNerd (System A)	• Risk Level: Medium • Regulation(s): Real Property Law § 226-b • Classification: Enforceable in most contexts • Explanation: This clause restricts the tenant's right to assign or sublet, which is regulated by law. The landlord's prior written consent is required, but the clause does not provide any further details or exceptions. • Suggested Improvement: Consider adding language to specify the conditions under which assignment or subletting is allowed or prohibited, such as "Tenant shall not assign the lease or sublet the premises without the prior written consent of the Landlord, which shall not be unreasonably withheld."	Specifying the allowance for service animals avoids any issues about the subject.
	GoHeather (System B)	• Risk Level: Medium • Explanation: This is a standard clause, but under New York law, tenants in buildings with four or more units have a statutory right to request subletting, and landlords cannot unreasonably withhold consent. The clause should reflect this statutory right. • Suggested Improvement: Replace "Tenant shall not assign the lease or sublet the premises without the landlord's prior written consent." with "Tenant shall not assign the lease or sublet the premises without the landlord's prior written consent, which shall not be unreasonably withheld in accordance with New York law."	
Tenant must provide written notice of intent to vacate at least 60 days prior to lease termination.	ContractNerd (System A)	• Risk Level: Medium • Regulation(s): General Obligations Law § 5-905 • Classification: Enforceable in most contexts • Explanation: This clause requires the tenant to provide written notice of intent to vacate, which is allowed by law. However, the required notice period may vary depending on the type of tenancy and the location. • Suggested Improvement: Consider adding language to specify the notice period required by law, such as "Tenant must provide written notice of intent to vacate at least [insert number] days prior to lease termination, as required by law."	Leaving the time period blank signals to the tenant that this is a time period that can and should be bargained for to address the circumstances of the individual landlord and tenant.

Table A3. *Cont.*

Clause	System	Analysis	Comment
	GoHeather (System B)	• Risk Level: Medium • Explanation: A 60-day notice period is longer than the standard 30 days in New York for most leases. This could be burdensome for the tenant and is not required by law unless specifically agreed. The tenant should negotiate for a 30-day notice period. • Suggested Change: Replace "Tenant must provide written notice of intent to vacate at least 60 days prior to lease termination." with "Tenant must provide written notice of intent to vacate at least 30 days prior to lease termination, unless otherwise required by law."	

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